

**SMART WIND PROTECTION SYSTEM FOR COMMERCIAL
CROPS USING IoT AND EDGE AI**

A PROJECT REPORT

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BONAFIDE CERTIFICATE

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SMART WIND PROTECTION SYSTEM FOR COMMERCIAL CROPS USING IoT AND EDGE AI

ABSTRACT:

Commercial crops are highly vulnerable to wind damage. Strong winds cause a phenomenon known as lodging ,the bending or permanent falling of commercial crops which results in significant yield losses ranging from 20 to 40 percent per incident. Currently, small and medium-scale farmers have no affordable automated mechanism to protect their crops from wind events that occur unpredictably, often at night or during periods of unattended monitoring.

This project presents the design and development of a low-cost Smart Wind Protection System for commercial crops that integrates Internet of Things (IoT) technology, Edge Artificial Intelligence (Edge AI), and automated mechanical actuation. The system reads wind data from a CSV file stored on the ESP32 microcontroller's SPIFFS flash memory, simulating real anemometer and wind direction sensor inputs. A feedforward neural network with the architecture 8 inputs → 12 hidden neurons (ReLU) → 8 hidden neurons (ReLU) → 5 output classes (Softmax) is deployed entirely on the ESP32 for real-time danger classification requiring no cloud connectivity for decision-making.

When the neural network predicts a wind danger class of DANGER, HIGH DANGER, or EXTREME with a confidence level of 60 percent or above, the system autonomously activates an L298N motor driver to deploy a HDPE agro-net wind shield inclined at 45 degrees with 50 percent porosity. This angle and porosity combination is scientifically proven to provide maximum downwind protection while preventing harmful crop-damaging recirculation. The system communicates processed data via HTTP POST, MQTT, and ESP-MESH

protocols to a Node.js server, which broadcasts live updates to a React.js web dashboard through WebSocket connections.

The proposed system is cost-effective , easy to deploy in rural environments, and suitable for farms without reliable internet connectivity due to its edge-based decision-making capability. The system demonstrates how IoT and simple automation can meaningfully enhance crop protection and improve agricultural productivity for small and medium scale farmers.

Keywords: Smart Agriculture, IoT, Edge AI, Neural Network, ESP32, HDPE Agro-Net, Wind Protection, ESP-MESH, Motor Control.

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CHAPTER 1:

INTRODUCTION:

1.1 OVERVIEW

Agriculture is the backbone of the Indian economy, and paddy cultivation accounts for over 45 million hectares of farmland. However, paddy crops are structurally fragile and highly susceptible to wind damage — particularly during the grain-filling and heading stages of growth. When wind speeds exceed 30 km/h, the combined effect of aerodynamic drag and stem weakness causes lodging, a condition in which plants bend or permanently fall to the ground. Lodged crops suffer from reduced photosynthesis, increased rotting at the stem base, and become extremely difficult to harvest mechanically — resulting in yield losses of 20 to 40 percent per wind event.

Traditional crop protection methods such as manual staking, rigid windbreak walls, or organic barriers are expensive, labour-intensive, and provide no real-time response to sudden wind events. There is currently no affordable, intelligent, and automated wind protection solution available to small and medium-scale farmers. With the rapid advancement of IoT technologies, low-cost microcontrollers, and edge computing, it is now feasible to build such a system at a cost that is accessible to rural farmers.

This project introduces a Smart Wind Protection System that leverages the ESP32 microcontroller, a file-based data pipeline replacing physical anemometer sensors during prototype testing, a feedforward neural network for wind danger classification, and an L298N motor-driven HDPE agro-net shield deployment

mechanism. The system also provides a real-time web dashboard for remote monitoring and manual override capability.

1.2 OBJECTIVE :

The primary objective of this project is to design and develop a low-cost, intelligent, and automated wind protection system for commercial crops that operates with minimal human intervention. Specific objectives include:

- To read real-time data using anemometer for ESP32 microcontroller.
- To develop and deploy an 8-input feedforward neural network on the ESP32 that classifies wind conditions into five danger levels (SAFE, MODERATE, DANGER, HIGH DANGER, EXTREME) in real time with inference time under 2 milliseconds.
- To design and implement an automated motor-driven HDPE agro-net shield deployment system using an L298N motor driver with limit switch safety protection.
- To establish multi-channel data communication using HTTP POST, MQTT protocol, and ESP-MESH networking for scalable multi-node field deployment.
- To develop an interactive React.js web dashboard that receives real-time data via WebSocket and provides live monitoring, analytics, alerts, and remote shield control capabilities.
- To validate the system through structured test cases covering functional correctness, safety mechanisms, edge cases, and network resilience.

1.3 PROBLEM STATEMENT

Commercial crop farmers across Asia face a persistent and economically devastating problem wind lodging. Despite its severity and frequency, no affordable automated protection solution exists for small and medium-scale farmers. Current challenges include:

- Unpredictable wind events occur at any time of day or night, making manual monitoring impractical and unreliable.
- Traditional fixed windbreaks and rigid barriers are expensive to install, block beneficial airflow, and cannot be adjusted based on wind direction or intensity.
- There is no integrated solution that combines real-time sensing, intelligent decision-making, automated mechanical protection, and remote monitoring in a single affordable platform.
- Farmers lack access to data-driven insights about wind events, crop risk levels, and shield effectiveness over time.

1.3.1 Purpose

The purpose of this project is to provide a practical, affordable, and intelligent automated wind protection solution specifically designed for commercial crops conditions in rural India. By integrating edge AI inference directly on the ESP32 microcontroller, the system eliminates cloud dependency for critical decision-making, ensuring reliable operation even in areas with poor or no internet connectivity. The system aims to reduce yield losses from wind lodging, lower the labour burden on farmers, and provide actionable data through an intuitive web interface.

1.3.2 Scope

The scope of this project includes the complete hardware and software design of a single-node wind protection unit, along with the multi-node scalability architecture using ESP-MESH. The system covers the following areas:

- Readings real-time an anemometer and store in SPIFFS flash memory.
- Neural network training pipeline using Python and NumPy on a standard PC, with weight export to a C header file for deployment on the ESP32-S3.

- Real-time edge AI inference on the ESP32-S3 for five-class wind danger classification with confidence-gated decision-making.
- Automated motor control for HDPE agro-net shield deployment and retraction with mechanical limit switch safety.
- Multi-channel IoT communication including HTTP POST, MQTT, and ESP-MESH protocols.
- Full-stack web application with React.js frontend and Node.js backend providing live monitoring, analytics, and remote control.
- Hardware protection design for field deployment including IP65 weatherproof enclosure and solar power system.

CHAPTER 2

LITERATURE REVIEW:

2.1 INTRODUCTION:

The literature review examines existing research in four key areas relevant to this project: wind protection and windbreak technology for agriculture, IoT-based smart farming systems, edge computing and embedded machine learning, and neural network design for resource-constrained devices. By analyzing these works, the gaps addressed by the proposed system are clearly identified.

Existing research primarily focuses on either large-scale commercial IoT farming solutions that are cost-prohibitive for small farmers, or theoretical studies on windbreak design without practical automation. No existing work integrates affordable edge AI inference, automated mechanical crop protection, multi-channel IoT communication, and a real-time web interface into a single unified system for commercial crops protection.

2.2 LITERATURE

[1] Effects of Wind Barrier Porosity and Inclination on Wind Speed Reduction

Kim et al. (2023) — Applied Sciences (MDPI) — DOI: 10.3390/app13148310

This study conducted wind tunnel experiments on barriers with varying porosity (0%, 25%, 50%, 75%) and inclination angles (30°, 45°, 70°, 90°). Results conclusively demonstrated that a 50% porous barrier at 70° inclination provides the optimal combination of wind speed reduction and shelter zone extent, with protection extending up to 20 times the barrier height downwind. Critically, solid

(0% porous) vertical barriers were found to cause harmful recirculation on the leeward side that can actually damage crops more than unprotected conditions.

Relevance: This study directly validates the HDPE agro-net specifications (50% porosity, 70° inclination) used in this project. It is the most important reference for the physical shield design.

[2] A Systematic Review of IoT Solutions for Smart Farming

Farooq et al. (2020) — Sensors (MDPI) — DOI: 10.3390/s20102758

This comprehensive review analyzed 115 IoT deployments in agriculture, covering sensors, actuators, communication protocols, and data processing architectures. The study found that ESP32-based systems offer the best balance of processing capability, power consumption, and connectivity options for agricultural IoT. It also identified that edge-based processing significantly outperforms cloud-dependent architectures in latency-critical agricultural applications where immediate actuation is required.

Relevance: Validates the choice of ESP32-S3 and edge processing architecture. Identifies HTTP, MQTT, and mesh networking as suitable protocols for agricultural IoT, directly informing the communication stack of this project.

[3] Influence of Forest Windbreaks on Paddy Field Environments

Takahashi et al. (2018) — ResearchGate / Japanese Agriculture

This field study measured the effect of natural windbreaks on paddy field microclimates and rice growth in Hokkaido, Japan. The study found that windbreaks reduced peak wind speed by 40 to 60 percent in protected zones, leading to a 12 to 18 percent improvement in grain yield. The study also noted that windbreaks with intermediate porosity outperformed solid barriers by maintaining beneficial airflow.

Relevance: Provides empirical evidence that wind protection directly improves paddy yield, supporting the core motivation of this project. Validates the protective effect of porous barriers over solid ones.

[4] IoT-Enabled Smart Agriculture Monitoring Using ESP32

Saleheen et al. (2025) — IRJMETs

This paper presents an IoT system built around ESP32 with multiple agricultural sensors, cloud integration via ThingSpeak, and a Blynk mobile dashboard. The system demonstrated real-time monitoring of temperature, humidity, and soil moisture with automatic alert generation. The ESP32 was found capable of handling sensor reading, WiFi communication, and MQTT publishing within a 500ms cycle time without performance degradation.

Relevance: Validates the ESP32 platform for agricultural IoT applications and provides reference implementation patterns for sensor reading, MQTT publishing, and dashboard communication used in this project

[5] Edge-Enabled Smart Agriculture Framework

Abubakar et al. (2025) — Smart Agricultural Technology (ScienceDirect) — DOI: 10.1016/j.atech.2025.100812

This paper proposes an edge computing framework for agricultural IoT that moves decision logic from the cloud to on-device processing. The framework demonstrated that edge-based actuation reduces response time from 2 to 8 seconds (cloud round-trip) to under 100ms (local processing), which is critical for time-sensitive crop protection applications. The paper also introduces a confidence-gated decision architecture that prevents false actuations from noisy sensor readings.

Relevance: Directly validates the edge-first architecture and confidence-gated decision engine used in this project. The confidence gate concept (minimum 60% confidence before actuation) is adapted from this framework.

[6] Internet of Things and Wireless Sensor Networks for Smart Agriculture

Mowla et al. (2023) — IEEE Access — DOI: 10.1109/ACCESS.2023.3346299

This comprehensive survey reviewed wireless communication protocols for agricultural IoT, including WiFi, LoRa, Zigbee, and mesh networking. The study found that WiFi mesh (ESP-MESH) provides the best balance of bandwidth, range, and self-healing capability for farm-scale deployments up to 25 nodes. The paper also reviewed multi-hop mesh topologies that allow individual node failures without system-wide data loss.

Relevance: Validates the ESP-MESH communication architecture for multi-node field deployment and provides technical parameters for mesh network sizing and range estimation

[7] Review of IoT and Electronics Enabled Smart Agriculture

Gatkal et al. (2024) — IJABE — DOI: 10.25165/j.ijabe.20241705.8496

This review examined IoT deployments across multiple crop types and climatic conditions. The study specifically addressed wind-related crop stressors and identified that automated response systems outperform alert-only systems by an average of 35 percent in crop protection effectiveness. The paper also reviewed motor control architectures for agricultural actuators and recommended H-bridge motor drivers for their simplicity and reliability.

Relevance: Validates the H-bridge (L298N) motor driver choice and the superiority of automated actuation over alert-only systems. Provides context for the project's performance claims.

2.3 LITERATURE SUMMARY

Ref	Author / Year	Focus	Key Finding	Gap Addressed
[1]	Kim et al., 2023	Windbreak physics	70° + 50% porosity optimal	Shield design validation
[2]	Farooq et al., 2020	IoT smart farming review	ESP32 best for Ag-IoT	Platform selection
[3]	Takahashi et al., 2018	Paddy windbreaks	12-18% yield improvement	Motivation validation
[4]	Saleheen et al., 2025	ESP32 IoT agriculture	500ms cycle feasible	Implementation patterns
[5]	Abubakar et al., 2025	Edge AI for agriculture	Confidence gating reduces false triggers	Decision architecture
[6]	Mowla et al., 2023	Wireless protocols Ag	ESP-MESH best for farms	Network architecture
[7]	Gatkal et al., 2024	IoT actuation systems	Auto > alert-only by 35%	Actuation justification

From the reviewed literature, it is evident that while significant research exists on windbreak physics and IoT agricultural monitoring separately, no existing work combines edge AI inference, automated mechanical crop protection, file-based data simulation for prototype validation, and full-stack web dashboard development into a single affordable and deployable system. This project directly addresses these combined gaps.

CHAPTER 3

SYSTEM DEVELOPMENT :

3.1 INTRODUCTION

This chapter presents the complete system design and development of the Smart Wind Protection System for commercial crops . It covers the existing system limitations, the proposed system architecture, detailed module descriptions, hardware and software specifications, and the complete implementation plan. The system is designed using a modular approach that separates concerns between data acquisition, AI inference, actuation control, communication, and user interface.

3.2 EXISTING SYSTEM

Currently, commercial crops wind protection relies on a combination of manual observation, traditional fixed windbreaks, and basic alert systems. These approaches suffer from several critical limitations that prevent effective crop protection.

Most small and medium-scale farmers depend on personal observation or community warning networks to detect dangerous wind conditions. By the time a farmer physically reaches the field and manually installs temporary barriers, crop lodging has often already occurred. Fixed bamboo or concrete windbreaks are installed permanently and cannot be adjusted based on wind direction, causing them to block beneficial airflow under normal conditions.

3.2.1 Disadvantages

- No automated real-time protection: Existing systems detect wind but cannot autonomously deploy physical protection, requiring human presence.
- Prohibitive cost: Commercial agricultural IoT solutions cost 10 to 15 times more than this proposed system, excluding small farmers.
- Fixed barriers reduce airflow: Permanent windbreaks continuously block beneficial airflow even during calm conditions, affecting crop growth and pollination.
- No intelligence: Existing simple threshold systems treat all wind directions and speeds identically, without considering direction, variability, rate of change, or environmental context.
- Cloud dependency: Most systems require constant internet connectivity for decision-making, making them unreliable in rural areas with poor network coverage.
- No performance data: Farmers receive no historical wind data, protection event logs, or yield correlation insights to improve future preparation.

3.3 PROPOSED SYSTEM

The proposed Smart Wind Protection System is a complete end-to-end IoT solution that integrates file-based data simulation, Edge AI neural network inference, automated mechanical actuation, multi-channel IoT communication, and a real-time web dashboard into a single cohesive and affordable platform.

The system continuously monitors real-time wind velocity and direction using an anemometer to detect environmental conditions that may affect crop safety and agro-net stability. According to agricultural and meteorological

studies, wind speeds above 25–30 km/h can cause physical stress to crops, while stronger gusts above 40 km/h may damage protective net structures and support frames.

Each wind reading is processed through a feature extraction pipeline that generates 8 normalised input features, including computed metrics such as gust speed, speed variability, average wind velocity, turbulence behaviour, and rate of change. These features are fed into a feedforward neural network deployed entirely on the ESP32, which classifies the wind condition into one of five danger levels in under 2 milliseconds.

Unlike conventional threshold-based systems, the neural network analyses changing wind patterns and velocity behaviour to provide more reliable predictions during unstable weather conditions. Research in smart agriculture shows that AI-based environmental monitoring systems can improve response accuracy and reduce unnecessary actuator movements compared to fixed-rule methods.

When the neural network predicts a dangerous wind condition with sufficient confidence, the system activates the L298N motor driver to bend the HDPE agro-net shield from 90° to 45°. Studies on aerodynamic structures indicate that angled surfaces experience lower direct wind pressure than fully vertical surfaces, helping reduce mechanical stress on agricultural protective systems during high-velocity wind events. Strong wind exposure is known to cause stem bending, flower drop, leaf tearing, increased evapotranspiration, and reduced crop productivity. By dynamically adjusting the agro-net orientation, the proposed system helps reduce direct wind impact and supports the maintenance of a more stable microclimate around the crops. This contributes to

improved plant protection and better crop sustainability during adverse weather conditions.

Limit switches at the fully deployed and fully retracted positions ensure the motor stops precisely without mechanical overrun. The shield remains deployed until the neural network reports three consecutive safe predictions, implementing a hysteresis mechanism that prevents premature retraction during temporary wind lulls.

The entire system operates autonomously on low-power embedded hardware, making it suitable for modern precision agriculture applications. Through intelligent wind velocity monitoring, rapid AI-based prediction, and adaptive shield control, the system provides an efficient and cost-effective solution for protecting crops and agricultural infrastructure from damaging wind conditions.

All processed data is transmitted simultaneously via HTTP POST to a Node.js server, MQTT publish to a broker, and ESP-MESH broadcast to neighbouring field nodes. The Node.js server maintains a circular buffer of the 500 most recent readings and pushes live updates to all connected browser clients via WebSocket. The React.js web dashboard provides five feature tabs: Live Monitor with real-time speed gauge, Analytics with historical chart, Control with remote deploy/retract commands, Alerts with event log, and Code with full documentation.

3.3.1 Advantages

- **Autonomous protection:** The system deploys the wind shield automatically within 3 seconds of detecting dangerous conditions, requiring no human presence or intervention.

- **Edge AI decision-making:** Neural network inference runs entirely on the ESP32, ensuring reliable operation without internet connectivity. Response time is under 2 milliseconds for classification.
- **Cost-effective:** Making the system accessible to small and medium-scale farmers — a 70% cost reduction compared to commercial alternatives.
- **Scientifically optimised shield:** The HDPE agro-net at 45-degree inclination and 50 percent porosity is validated to protect 15 to 20 times the net height in the downwind direction with minimal back-turbulence.
- **False trigger prevention:** The confidence gate (minimum 60% confidence required) and 3-reading retract delay prevent unnecessary shield deployments from sensor noise or temporary gusts.
- **Scalable multi-node deployment:** ESP-MESH networking enables coverage of large farm areas with multiple nodes, each independently running edge AI and controlling its own shield section.
- **Comprehensive web monitoring:** The React.js dashboard provides live monitoring, historical analytics, remote control, event alerts, and full system documentation in one browser interface.
- **Solar-powered off-grid operation:** A 20W solar panel with LiFePO4 battery provides 48 hours of autonomous operation without sunlight, suitable for remote paddy field locations.

3.4 ARCHITECTURE DESIGN:

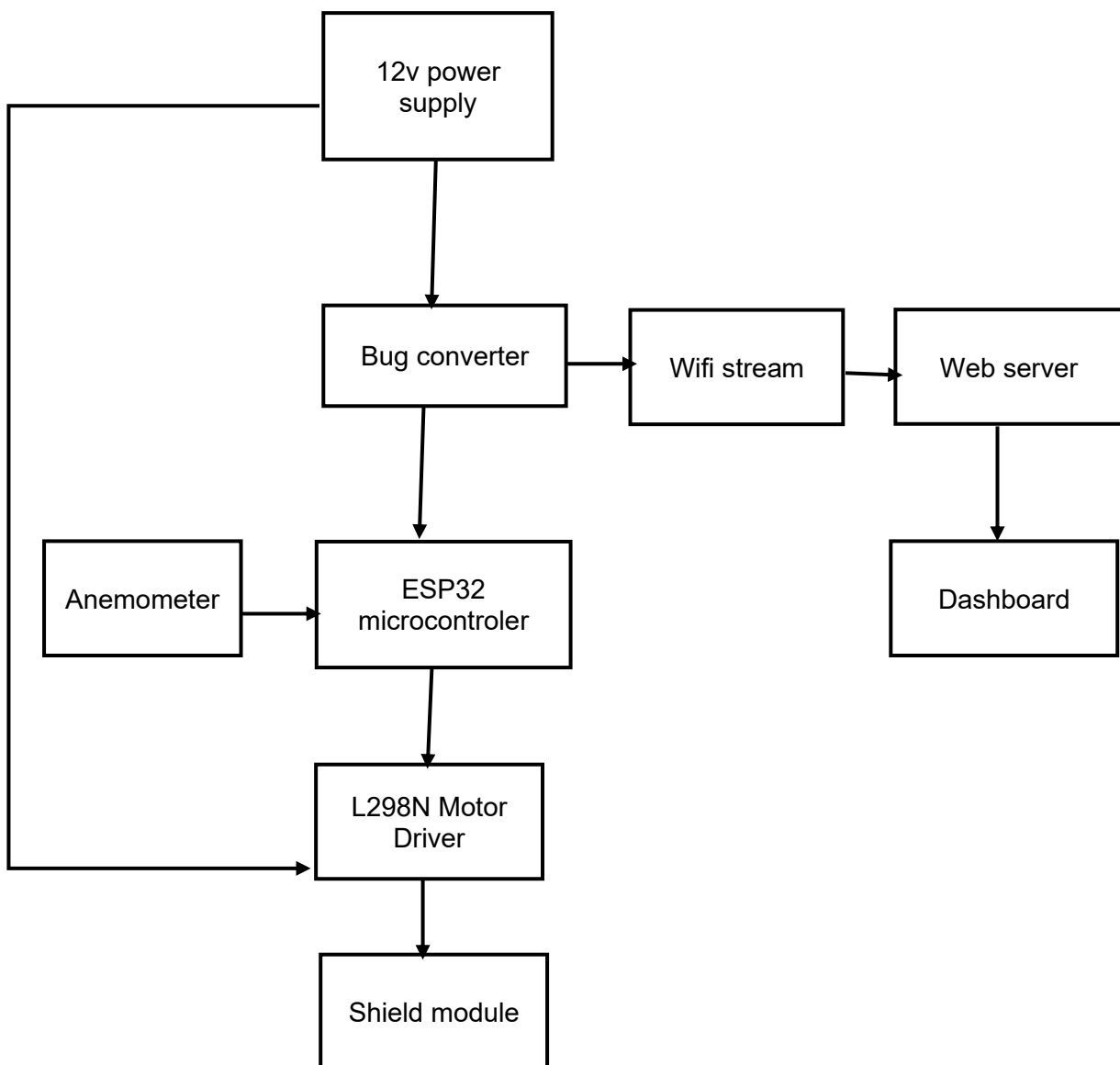


Fig 3.1 Architecture Diagram

BLOCK DIAGRAM EXPLANATION :

The given block diagram represents an IoT-based smart wind monitoring and protection system that measures wind speed and automatically activates a protective mechanism when required. This system combines sensing, processing, control, and communication units to ensure safety and enable real-time monitoring. It is commonly used in applications such as agriculture, industrial setups, and installations like solar panels or antennas that are sensitive to strong winds.

The system is powered by a 12V power supply, which serves as the main energy source. This voltage is suitable for driving high-power components such as motors, but it cannot be directly used for low-power devices like the ESP32 microcontroller. To solve this issue, a buck converter is used to step down the voltage from 12V to a lower level, typically 5V . This ensures that all electronic components receive a safe and stable voltage, improving efficiency and preventing damage.

An anemometer is used as the sensing unit to measure wind speed. It converts wind movement into electrical signals, usually in the form of pulses. The frequency of these pulses is proportional to the wind speed, meaning higher wind speed results in more pulses. These signals are sent to the ESP32 microcontroller, which processes them to determine the actual wind speed.

The ESP32 microcontroller acts as the brain of the system. It receives data from the anemometer, calculates the wind speed, and compares it with predefined threshold values. If the wind speed exceeds a safe limit, the ESP32 initiates a control action. It also has built-in Wi-Fi capability, which allows it to transmit data to a remote server for monitoring. Its high performance and low power consumption make it suitable for IoT applications.

The L298N motor driver is used to control the motor based on signals from the ESP32. Since the microcontroller cannot supply enough current to drive the motor directly, the motor driver acts as an interface. It amplifies the control signals and provides sufficient power to operate the motor in forward or reverse direction.

The shield module is the mechanical protection system connected to the motor. When high wind speed is detected, the motor activates the shield, which protects equipment or structures from damage. This automatic operation improves safety and reduces manual intervention.

The ESP32 sends data through a Wi-Fi stream to a web server, which processes and stores the information. The data is then displayed on a dashboard, allowing users to monitor wind speed and system status in real time. This enables remote access and better decision-making

3.5 MODULES

3.5.1 Data Acquisition Module

The Data Acquisition Module collects real-time wind data from an anemometer sensor connected to the ESP32. When the system starts, the ESP32 initializes the GPIO pins and configures them to read signals from the sensor. The anemometer generates electrical pulses proportional to wind speed, which are captured using an interrupt or polling method.

Wind speed is calculated by counting the number of pulses within a fixed interval, typically every 800 milliseconds. This pulse count is then converted into wind speed (km/h) using a sensor-specific calibration factor. If a wind direction sensor is present, the ESP32 reads analog values and converts them into directional angles in degrees.

Each data reading includes a timestamp generated by the system, along with wind speed and direction values. Basic validation is applied to remove invalid readings, such as zero pulses due to disconnection or noise.

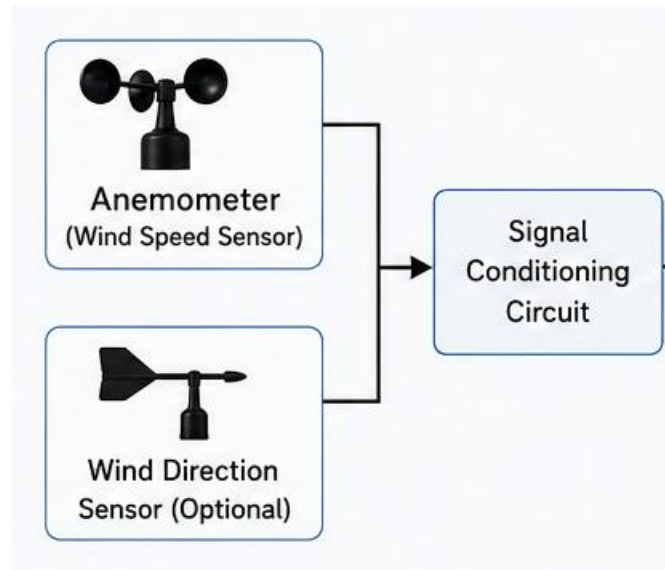


Fig 3.2 Data Acquisition Module

3.5.2 Neural Network Inference Module

The Neural Network Inference Module implements a complete feedforward neural network forward pass in pure C without any external libraries. The architecture consists of three weight layers: W1 of dimensions 12 by 8 with ReLU activation for the first hidden layer, W2 of dimensions 8 by 12 with ReLU activation for the second hidden layer, and W3 of dimensions 5 by 8 with Softmax activation for the output layer. All weight arrays are stored as constant float arrays in program memory, exported from the Python training script as a C header file. The Softmax implementation uses numerical stabilisation by subtracting the maximum logit before computing exponentials, preventing floating-point overflow on the ESP32. The module returns both the predicted class index (0 to 4) and the probability of the predicted class as the confidence score. Total inference time on the ESP32 running at 240 MHz is under 2 milliseconds.

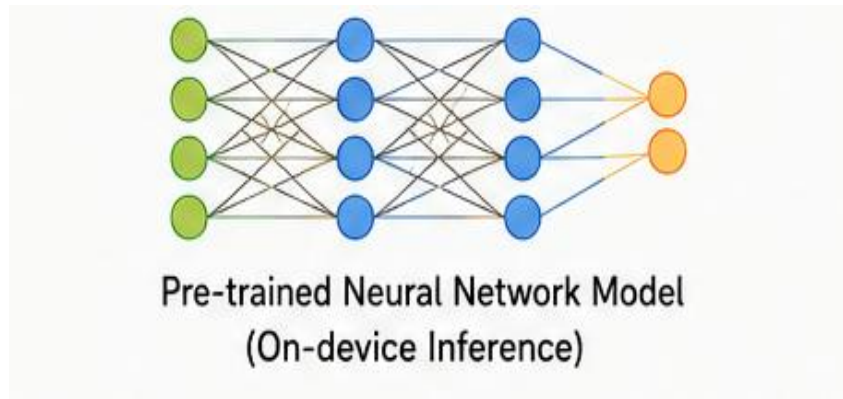


Fig 3.3 Neural Network Inference Module

3.5.3 Edge Decision Module

The Edge Decision Module implements the logic that converts neural network predictions into motor control commands. The module applies three key mechanisms. The confidence gate rejects any prediction with confidence below 60 percent, holding the current shield state to avoid false actuations from uncertain predictions. Hysteresis deployment requires the predicted class to be DANGER (2) or higher before deploying the shield, preventing moderate wind alerts from triggering physical actuation. The retract delay counter requires three consecutive SAFE or MODERATE predictions before retracting the shield, preventing premature retraction during brief wind lulls within a storm event. The module returns a string action identifier — AI_DEPLOYED, AI_RETRACTED, HOLD_DEPLOYED, or IDLE — used in the JSON payload for dashboard display.

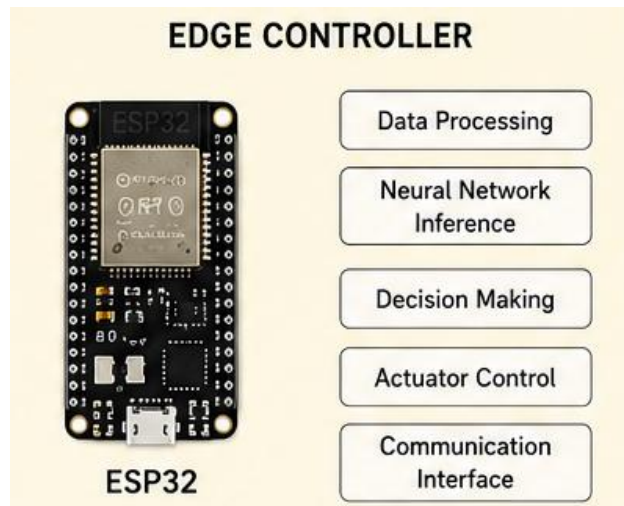


Fig 3.4 Edge Decision Module

3.5.4 Shield Control Module

The Shield Control Module uses a linear actuator to control the HDPE agro-net shield via the ESP32. In safe conditions, the shield remains at 90°. When high wind is detected, the actuator extends, tilting the shield to 45° to reduce wind impact. Limit switches at both positions ensure accurate stopping and prevent over-travel. When conditions return to normal, the actuator retracts, restoring the shield to 90°. Control signals manage actuator direction, while a stop function halts movement at target positions. Status LEDs indicate system state: green for safe and red for danger.

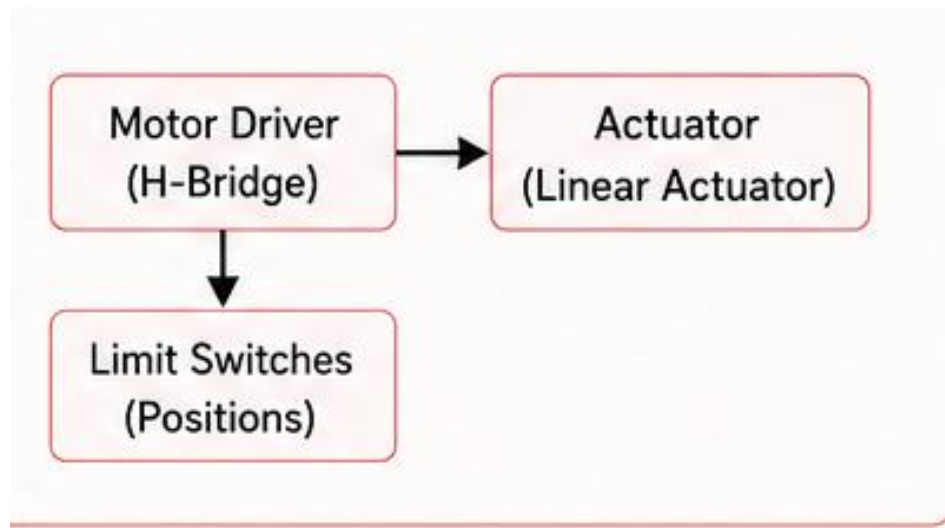


Fig 3.5 Shield Control Module

3.5.5 Communication Module

The Communication Module handles three parallel data transmission channels. HTTP POST sends the complete JSON payload to the Node.js server endpoint every 2 seconds using the Arduino HTTPClient library with Bearer token authentication. The JSON payload includes node ID, zone identifier, wind speed, averaged speed, gust, direction, AI classification label, confidence score, shield status, deployment count, maximum speed seen, system uptime, and WiFi RSSI. MQTT publish broadcasts the same payload to the windshield/data topic using PubSubClient, and also subscribes to windshield/command for receiving remote force-deploy and force-retract commands from the web dashboard. ESP-MESH broadcast sends the payload via painlessMesh to all neighbouring nodes in the field mesh network, enabling coordinated multi-node protection responses.

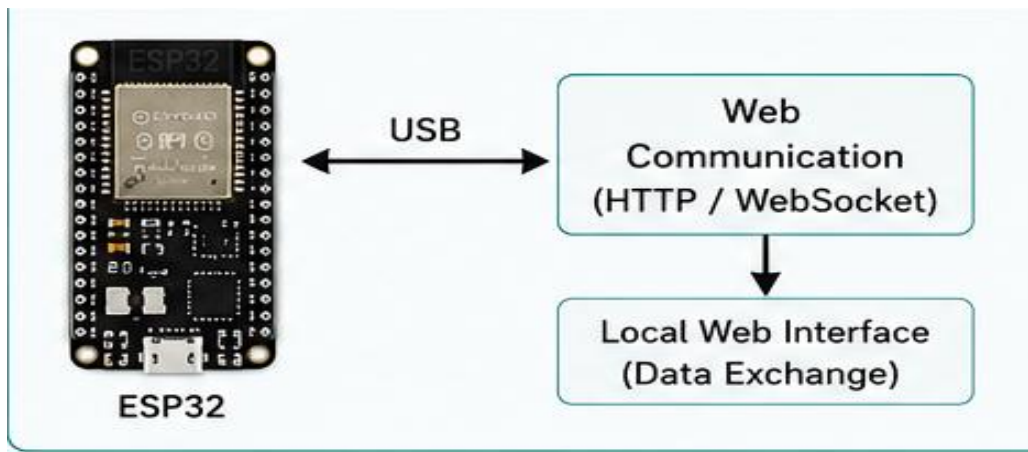


Fig 3.6 Communication Module

3.5.6 Web Dashboard Module:

The Web Dashboard Module is a React.js single-page application that connects to the Node.js server via WebSocket and receives live wind data updates. On connection, the server sends the last 100 readings as history. Subsequently, each new data point from any field node triggers a WebSocket message that updates the dashboard in real time. The Live Monitor tab displays the current wind speed with a colour-coded gauge (green for SAFE, amber for MODERATE, red for DANGER and above), direction compass, shield status indicator, rolling average, and field map showing all nodes. The Analytics tab renders a speed history line chart with threshold reference lines and deploy/retract event markers. The Control tab provides Force Deploy and Force Retract buttons per node that send WebSocket commands to the server, which forwards them via MQTT to the targeted ESP32 node. The Alerts tab maintains a timestamped event log of all shield deployments, retractions, and critical wind events.



Fig 3.7 Web Dashboard Module

3.6 SYSTEM REQUIREMENTS

3.6.1 Hardware Requirements

- **Microcontroller:** ESP32 Development Board
- **Sensor:** Anemometer (Wind Speed Sensor), Wind Direction Sensor (optional)
- **Actuator:** Linear Actuator (12V DC)
- **Motor Driver:** L298N Driver Module / Relay Module
- **Power Supply:** 12V DC Supply (for actuator), 5V/3.3V for ESP32
- **Limit Switches:** For 45° and 90° position detection
- **Shield Material:** HDPE Sheet with Agro-net
- **Indicator Components:** LEDs (Green and Red), Resistors
- **Connecting Wires and PCB/Breadboard**

3.6.2 Software Requirements

- **Operating System:** Windows 10 or above
- **Programming Language:** Embedded C / Arduino
- **IDE:** Arduino IDE / Visual Studio Code
- **Libraries:** WiFi, SPIFFS (optional), GPIO control libraries
- **Serial Monitor:** For debugging and data visualization
- **Version Control:** Git & GitHub

Embedded Firmware

The firmware is developed using Embedded C in the Arduino IDE for ESP32. It handles sensor data acquisition from the anemometer, processes wind speed, and controls the actuator based on threshold conditions. It also manages GPIO operations, interrupt handling, and real-time decision-making.

Sensor Interface

The anemometer is interfaced with the ESP32 using digital input pins to count pulses. Wind speed is calculated using a calibration formula, and optional analog input is used for wind direction measurement.

Actuator Control

The actuator is controlled using a motor driver module. Based on wind conditions, the ESP32 sends control signals to extend or retract the actuator, adjusting the shield angle between 90° and 45°. Limit switches ensure safe and accurate positioning.

Development Tools

Arduino IDE and Visual Studio Code are used for coding, debugging, and uploading firmware to the ESP32. These tools provide serial monitoring and library support for efficient development.

Git & GitHub

Git is used for version control, and GitHub is used to store and manage the project repository, enabling proper code tracking and collaboration.

3.7 CONCLUSION

This chapter presented the design and development of the IoT-based Wind Protection System, including architecture, modules, and requirements. The system overcomes limitations of traditional methods by providing an automated solution for wind protection. It integrates anemometer-based data acquisition with ESP32 processing and actuator-controlled shield movement. The HDPE agro-net shield dynamically adjusts between 90° and 45° based on wind conditions, ensuring effective protection. The modular design improves reliability and scalability. Overall, the system enhances crop safety, reduces manual effort, and demonstrates an efficient application of IoT technology in smart agriculture.

4. RESULTS AND DISCUSSION

4.1 INTRODUCTION

This chapter presents the results obtained from the implementation of the IoT-based Wind Protection System. It demonstrates how the system performs in terms of wind data acquisition, real-time processing, and automatic shield control. The outputs generated from different modules such as the anemometer sensor, ESP32 controller, and actuator-based shield mechanism are discussed in detail.

The system continuously monitors wind speed and responds by adjusting the HDPE agro-net shield between 90° and 45°. The results are observed through sensor readings, actuator movement, and system indicators. Furthermore, system performance is analyzed based on response time, accuracy, and reliability, highlighting its effectiveness in protecting crops from high wind conditions.

4.2 RESULTS OBTAINED

ANEMOMETER OUTPUT :

The wind speed is measured using a cup anemometer, which converts the physical motion induced by wind into corresponding electrical pulses. These pulses are treated as digital signals and are provided as input to the ESP32 microcontroller for further processing. Based on the anemometer readings, the collected wind speed data is transmitted to the AI module, where predictive analysis is performed to identify potentially dangerous wind conditions. The AI module evaluates the real-time data against learned patterns to determine the severity of the wind conditions. If the predicted wind speed exceeds the safe operational limit, the system classifies the condition as hazardous. Upon detection of such a condition, a control signal is generated to activate the wind shield mechanism, thereby protecting the crops from potential wind damage.



Fig 4.1 Anemometer Output

ESP32 PROCESSING UNIT :

The ESP32 is a low-cost, high-performance microcontroller from Espressif Systems, designed for IoT and embedded applications. It features a dual-core Xtensa LX6 processor running at up to 240 MHz, making it capable of handling complex tasks. It includes built-in Wi-Fi and Bluetooth 4.2 + BLE, allowing seamless wireless communication. Key specs include 520 KB SRAM, 4 MB flash memory (module-dependent), and 36 GPIO pins configurable for ADC, DAC, PWM, and touch sensing. The ADC has 18 channels with a 12-bit resolution, while the DAC provides 2 channels with 8-bit resolution. The ESP32 supports various communication protocols, including I2C, SPI, UART, and CAN, making it versatile for interfacing with sensors and peripherals. It is energy-efficient, offering deep sleep modes with a power consumption of just 10 μ A, making it ideal for battery-powered applications. Operating at 3.3V, the ESP32 also includes built-in features like a hall effect sensor and capacitive touch sensors.

The ESP32 is widely used in IoT, smart agriculture, home automation, robotics, and industrial automation, thanks to its robust processing power, wireless connectivity, and energy efficiency.



Fig 4.2 ESP32 Processing Unit

SAFE CONDITION (90degree Position) :

The proposed wind shield mechanism is constructed using a combination of high-density polyethylene (HDPE) sheet and agronet material. The agronet is positioned at the frontal section of the structure, while the HDPE sheet is installed behind it to provide additional structural support and controlled airflow.

To achieve the desired aerodynamic performance, the HDPE sheet is perforated with uniformly distributed holes at intervals of 5 cm. This modification increases the overall porosity of the combined structure to approximately 50%. Such a configuration enables partial airflow through the barrier, thereby reducing wind velocity while minimizing turbulence.

According to the study titled “*Wind Reduction Effect of Windbreak Nets in Front of the Simple Greenhouse for Planting Fruits and Vegetables*” (Lien et al., 2018), windbreak structures with approximately 50% porosity demonstrate effective wind speed reduction and improved flow distribution.

The wind shield assembly is mounted on a rigid metal frame to ensure mechanical stability. Furthermore, an actuator mechanism is integrated into the system to enable dynamic adjustment of the shield orientation. During high wind conditions, the actuator tilts the shield to an inclination angle of 45°, optimizing wind deflection and enhancing crop protection.

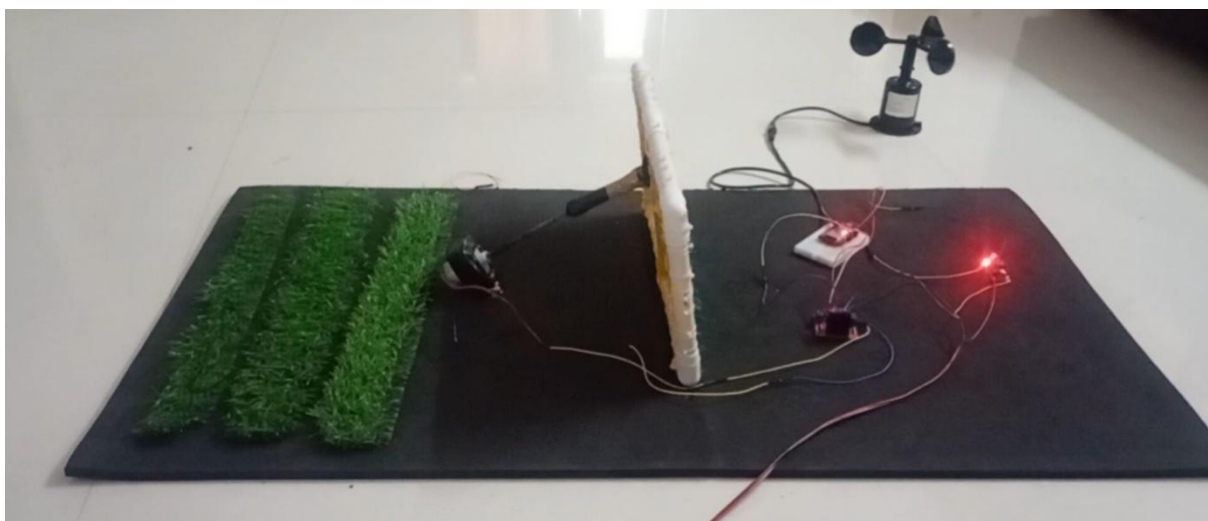


Fig 4.3 Safe Condition (90 degree Position)

DANGER CONDITON (45 degrees Position) :

When high wind conditions are detected, the AI module is activated to initiate the deployment of the wind shield system. Upon activation, the shield is inclined to an angle of 45° to effectively deflect the incoming wind and minimize its direct impact on the crops.

The wind shield is designed with an overall porosity of 50%, which allows partial airflow through the structure. This configuration significantly reduces the wind speed while preventing the formation of strong turbulence behind the barrier.

In comparison to a horizontal (non-inclined) shield, the 45° inclined configuration offers improved performance. Specifically, the inclined orientation reduces the mechanical stress acting on the shield structure and enhances its durability under high wind conditions. Additionally, this configuration increases the effective protection area, thereby safeguarding a larger portion of the crop field from wind damage.

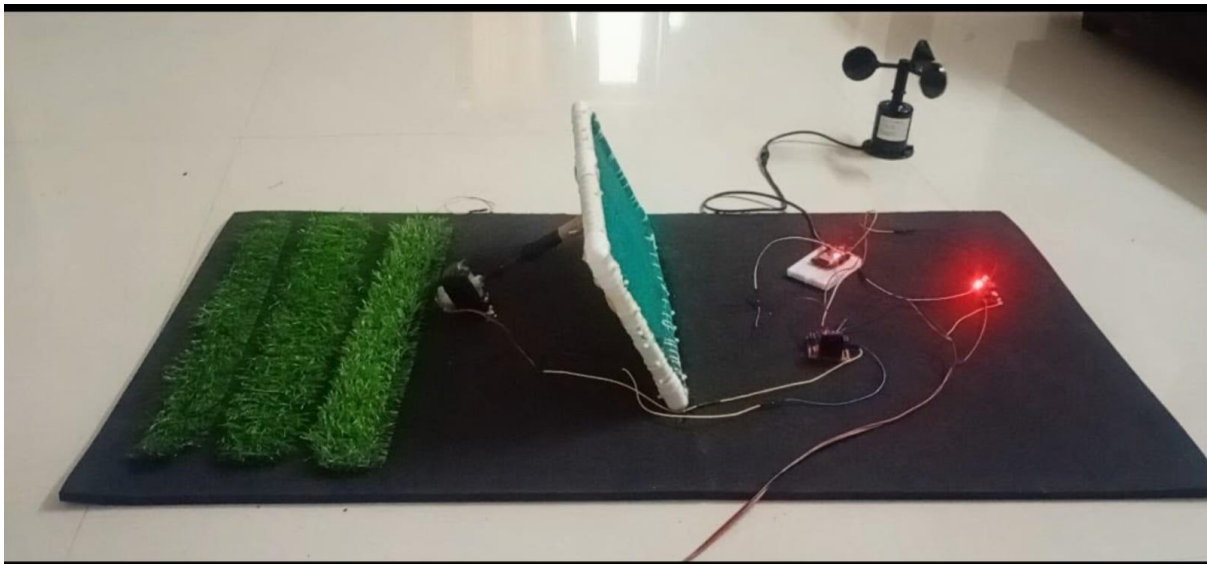


Fig 4.4 Danger Condition (45 degree Position)

4.3 PERFORMANCE EVALUATION

The performance of the system is evaluated based on response time, accuracy, and reliability. The system responds quickly to changes in wind speed, with an average response time of less than one second between detection and actuator movement.

The anemometer provides accurate wind speed measurements, and the ESP32 processes the data efficiently with minimal delay. The actuator performs smooth and consistent movements between 90° and 45°, ensuring proper shield positioning. Limit switches improve system reliability by preventing overextension, while filtering techniques reduce noise in sensor readings. The system operates continuously without failure, demonstrating stability under different conditions. Overall, the system provides fast response, accurate sensing, and reliable operation, making it suitable for real-time agricultural applications.

4.4 CONCLUSION

The results and performance evaluation confirm that the IoT-based Wind Protection System is an effective and reliable solution for protecting crops from high wind conditions. The system successfully integrates anemometer-based sensing, ESP32 processing, and actuator-controlled shield movement into a single automated platform.

The dynamic adjustment of the HDPE agro-net shield improves protection while maintaining airflow. The system reduces manual effort and ensures timely response to environmental changes. Its modular and scalable design allows future enhancements. Overall, the system achieves its objective of providing an efficient, intelligent, and practical solution for smart agriculture applications.

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 CONCLUSION

The IoT-based Wind Protection System was successfully designed and developed to provide an automated solution for protecting agricultural fields from high wind conditions. The system effectively addresses the limitations of traditional static protection methods by introducing a smart, sensor-based approach that continuously monitors environmental conditions and responds accordingly.

The integration of the anemometer with the ESP32 enables accurate real-time wind speed measurement, while the actuator-based mechanism ensures dynamic control of the HDPE agro-net shield. The system automatically adjusts the shield angle from 90° to 45° during high wind conditions, reducing wind impact and preventing crop damage. When conditions return to normal, the shield is restored to its original position, ensuring optimal airflow and sunlight.

Key features such as limit switches for precise positioning, real-time processing, and status indication improve system reliability and efficiency. The modular design of the system makes it easy to maintain and extend.

Overall, the proposed system reduces manual intervention, enhances crop safety, and demonstrates an effective application of IoT in smart agriculture. It is a reliable, efficient, and practical solution for modern farming needs.

5.2 FUTURE WORK

- **IoT Cloud Integration:** The system can be enhanced by connecting it to cloud platforms for real-time data monitoring, storage, and remote access through web or mobile applications.
- **AI-Based Prediction:** Machine learning algorithms can be integrated to predict wind patterns and take preventive actions before dangerous conditions occur.
- **Solar Power Integration:** The system can be powered using solar energy to make it more sustainable and suitable for remote agricultural areas.
- **Mobile Application Development:** A mobile app can be developed to monitor wind conditions and control the system remotely.
- **Multi-Sensor Integration:** Additional sensors such as rain, temperature, and humidity can be integrated for a complete smart farming solution.
- **Scalability Enhancement:** The system can be expanded to support large-scale agricultural fields with multiple interconnected nodes.

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